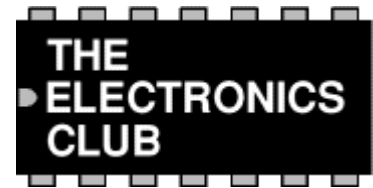


[Home](#) | [Map](#) | [Projects](#) | [Construction](#) | [Soldering](#) | [Study](#) | [Components](#) | [555](#) | [Symbols](#) | [FAQ](#) | [Links](#)

Transducers

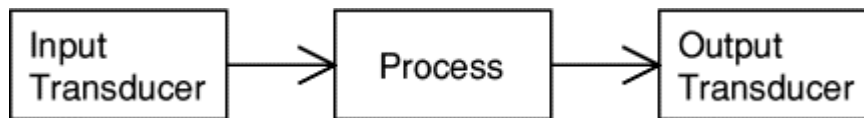


[Input Transducers](#) | [Output Transducers](#) | [Using Input Transducers](#)

Next Page: [Voltage Dividers](#)

A transducer is a device which converts a signal from one form to another.

Most electronics circuits use both input and output transducers:



Input Transducers

Input Transducers convert a quantity to an electrical signal (voltage) or to resistance (which can be converted to voltage). Input transducers are also called **sensors**.

Examples:

- [LDR](#) converts brightness (of light) to resistance.
- [Thermistor](#) converts temperature to resistance.
- Microphone converts sound to voltage.
- [Variable resistor](#) converts position (angle) to resistance.



LDR

Photograph © [Rapid Electronics](#)

Output Transducers

Output Transducers convert an electrical signal to another quantity.

Examples:

- [Lamp](#) converts electricity to light.
- [LED](#) converts electricity to light.
- [Loudspeaker](#) converts electricity to sound.
- Motor converts electricity to motion.
- Heater converts electricity to heat.



Loudspeaker

Photograph © [Rapid Electronics](#)

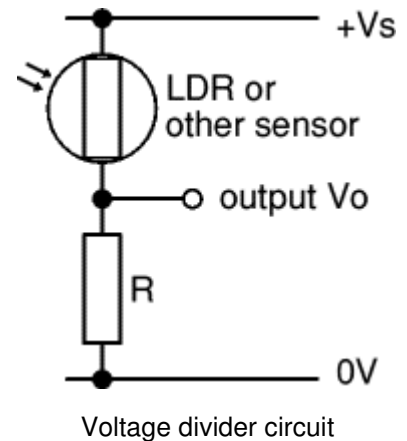
Using input transducers (sensors)

Most input transducers (sensors) vary their **resistance** and this can be used directly in some circuits but it is usually converted to an electrical signal in the form of a **voltage**.

The voltage signal can be fed to other parts of the circuit, such as the input to a chip or a transistor switch.

The conversion of varying resistance to varying voltage is performed by a simple circuit called a **voltage divider**.

Please see the next page for more information about [voltage dividers](#).



Next Page: [Voltage Dividers](#) | [Studying Electronics](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
- [Soldering Guide](#)
- [Study Electronics](#)
- [Electronic Components](#)
- [555 Timer](#)
- [Circuit Symbols](#)
- [Frequently Asked Questions](#)
- [Links to other Electronics sites](#)



© John Hewes 2007, The Electronics Club, www.kpsec.freeuk.com